



INTRODUCTION TO RESEARCH METHODOLOGY



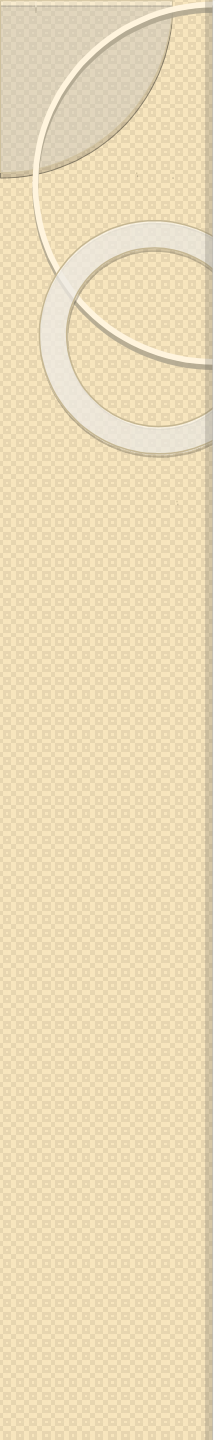
OVERVIEW

- Meaning of Research
- Objectives
- Importance
- Research Methods Vs Methodology
- Motivation for research
- Qualities of a good research
- Types of research

MEANING OF RESEARCH

The systematic method consisting of

- defining the problem,
- formulating a hypothesis,
- collecting the facts or data,
- analysing the facts and
- reaching certain conclusion either
 - in the form of solutions towards the concerned problem or
 - in certain generalisation for some theoretical formulation



AIM OF RESEARCH

- To find out truth which is hidden and which has not been discovered yet

OBJECTIVES OF RESEARCH

- To gain familiarity with a phenomenon or to achieve new insights into it – *Exploratory or formulative studies*
- To portray accurately the characteristics of a particular individual, situation or a group – *Descriptive studies*
- To determine the frequency with which something occurs or with which it is associated with something else – *Diagnostic Studies*
- To test a hypothesis of a casual relationship between variables – *Hypothesis-testing studies*



IMPORTANCE OF RESEARCH

Research is important, as it:

- Facilitates decision making in all spheres of business and industry
- Is useful in studying social problems and seeking solutions to these problems
- Is a vital tool in several fields of economic and government policy formulation

So in a nut shell, *research* unveils a fountain of knowledge which is a major drive in the development and well being of mankind.



RESEARCH METHOD Vs METHODOLOGY

Methods - the methods/techniques the researchers use in performing research operations.

It includes:

- The methods used for collecting the data
- The statistical or analytical tools and techniques used to analyse the data or to establish relationships b/w data and unknowns
- The methods used to measure the accuracy of the results obtained



RESEARCH METHOD Vs METHODOLOGY

Methodology is a systematic way to solve the research problem.

- Science of understanding how research is done
- Study various steps adopted by a researcher
- Researchers should know the relevant method and which are not



RESEARCH METHOD Vs METHODOLOGY

When we talk of research methodology

- we not only talk of the research methods but also consider the logic behind the methods we use in the context of our research study and
- explain why we are using a particular method or technique and
- why we are not using others so that research results are capable of being evaluated either by the researcher himself or by others



MOTIVATION FOR RESEARCH

- Desire to get a research degree along with its consequential benefits;
- Desire to face the challenge in solving the unsolved problems, i.e., concern over practical problems initiates research;
- Desire to get intellectual joy of doing some creative work;
- Desire to be of service to society;
- Desire to get respectability.



MOTIVATION FOR RESEARCH

Classifications – 4 types

- i. Intrinsic motivation
- ii. Extrinsic motivation
- iii. Personal motivation
- iv. Interpersonal motivation



INTRINSIC MOTIVATION

- Love of the work itself
- To get the joy of doing some creative work
- Curiosity to find unknown facts
- Curiosity to know new things
- It includes interest, challenge, creative flow, learning, meaning and purpose
- High levels of intrinsic motivation leads to
OUTSTANDING CREATIVE PERFORMANCE



EXTRINSIC MOTIVATION

- Rewards for good work
- Punishments for poor work
- Money, fame
- Awards, praise, status, opportunities
- Deadlines, commitments, bribes, threats
- Too much forces on extrinsic motivation blocks
CREATIVITY



PERSONAL MOTIVATION

- Authenticity, generosity, public recognition, knowledge, pleasure
- To serve society by solving social problems
- To acquire respectability



INTERPERSONAL MOTIVATION

- Influences from other people
- Response to people around us (copying, competition, rebellion, encouragement)



QUALITIES OF A GOOD RESEARCH

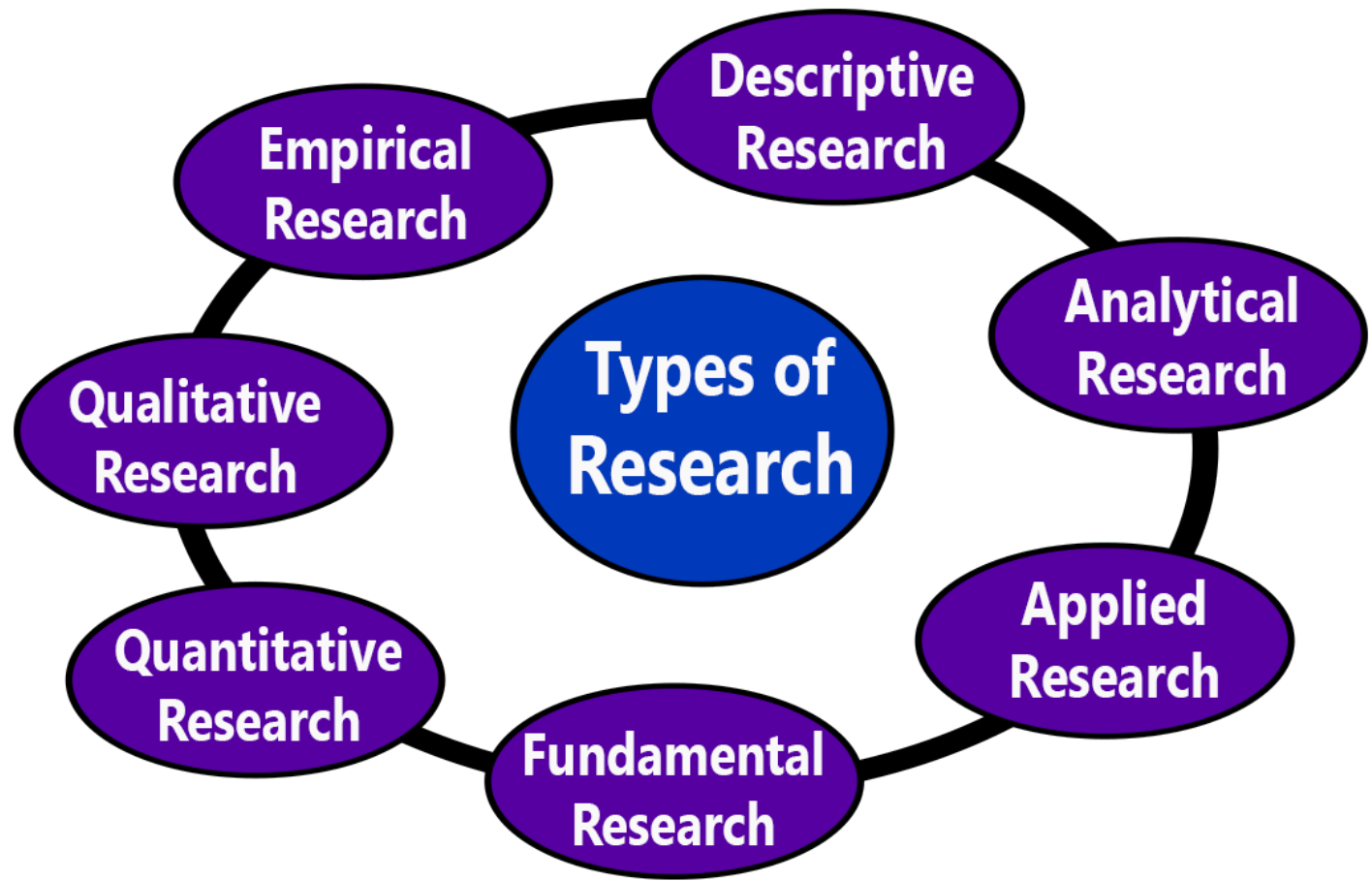
- *Systematic* – follow logical sequence as per well defined set of rules
- *Logical* – always relay on logical reasoning. This features makes research meaningful and appealing
- *Empirical* – deals with real situation and proves its validity with concrete foundation
- *Replicable* – results can be verified by replicating the study



TYPES OF RESEARCH

1. Descriptive Research
2. Analytical Research
3. Applied Research
4. Quantitative Research
5. Qualitative Research
6. Conceptual Research
7. Empirical Research
8. Other Types of Research

TYPES OF RESEARCH



DESCRIPTIVE RESEARCH

- Purpose is description of a state of affairs as it exists at present
- Includes surveys and fact-finding enquiries of different kinds
- Also known as *Ex post facts research*
- Characterised by the fact that the research can only report what has happened or what is happening and has no control over the variables
- Eg: A study to unveil the income level of people in rural India, their shopping habits, brand preferences etc. - Survey methods



ANALYTICAL RESEARCH

- The researcher has to use facts or information already available, and
- Analyse these to make a critical evaluation of the material



APPLIED RESEARCH

- Aims at finding a solution for an immediate problem facing a society/an industrial/business organisation
- Central aim is to discover a solution for some pressing practical problems
- Eg: Research to identify social, economic or political trends that may affect a particular institution, marketing research, evaluation research etc.



FUNDAMENTAL RESEARCH

- Concerned with generalisations and with the formulation of a theory
- “Gathering knowledge for knowledge’s sake” is fundamental or pure or basic research
- Directed towards finding information that has a broad base of applications
- Eg: Research concerning some natural phenomenon or relating to pure mathematics, research studies etc.
- Eg: Research concerning human behaviour carried on with a view to make generalisations about human behaviour



QUANTITATIVE RESEARCH

- Based on the quantitative measurements of some characteristics.
- Applicable to phenomena that can be expressed in terms of quantity.
- It depends on numerical data, such as statistics and measurements, to investigate specific questions, like who, what, where or when.
- The results are usually presented in tables or graphs.



QUANTITATIVE RESEARCH

- Example: A car manufacturer compares the number of sales of red sedans compared to white sedans. The research uses objective data—the sales figures for red and white sedans—to draw conclusions.

QUALITATIVE RESEARCH

- Concerned with qualitative phenomenon - phenomena relating to or involving quality or kind
- Motivation Research – an important type - investigating the reasons for human behaviour (i.e., why people think or do certain things)
- Aims at discovering the underlying motives and desires, using in depth interviews for the purpose
- Other techniques are word association test, story completion test etc.
- Attitude or opinion research – designed to find out how people feel or what they think about a particular subject is also a qualitative research



QUALITATIVE RESEARCH

- Specially important in behavioural sciences – aims to discover the underlying motives of human behaviour
- Seek guidance from experimental psychologists to apply qualitative research in practice



CONCEPTUAL RESEARCH

- Related to some abstract ideas or theory
- Generally used by philosophers and thinkers to develop new concepts or to reinterpret existing ones



EMPIRICAL RESEARCH

- Relies on experience or observations alone, often without due regard for system and theory
- It is Data-based research, coming up with conclusions which are capable of being verified by observation or experiment
- Also known as experimental type of research
- Works to get enough facts to prove or disprove hypothesis
- Evidence gathered by this research is considered to be the most powerful support possible for a given hypothesis

SOME OTHER TYPES OF RESEARCH

- One-time Research
 - Research confined to a single time period
- Longitudinal Research
 - Carried over several time periods
- Field-setting /Laboratory /Simulation Research
 - Depends upon the environment in which it is to be carried out
- Clinical or Diagnostic Research
 - Follows case study methods or indepth approaches to reach the basic causal relations

SOME OTHER TYPES OF RESEARCH

- go deep into the causes of things or events that interest us, using very small samples
- Historical Research
 - Utilizes historical sources like documents, remains etc.
- Conclusion-oriented Research
 - Free to pick up a problem, redesign the enquiry as he proceeds and is prepared to conceptualize as he wishes
- Decision-oriented Research
 - Always for the need of a decision maker

Research Process

- Research process consists of a series of steps or actions required for effectively conducting research.
- The following are the steps that provide useful procedural guidelines regarding the conduct of research:
 - (1) Formulating the research problem;
 - (2) Extensive literature survey;
 - (3) Developing hypothesis;
 - (4) Preparing the research design;
 - (5) Determining sample design;
 - (6) Collecting data;
 - (7) Execution of the project;
 - (8) Analysis of data;
 - (9) Hypothesis testing;
 - (10) Generalization and interpretation
 - (11) Preparation of the report or presentation of the results - the formal write-up of conclusions reached.

Thinking Skills

- Critical thinking skills are the mental process involved in processing information.
- They help us with problem solving, decision making, and thinking critically.



4 Types of Thinking Skills

1. Convergent Analytical Thinking

- Convergent thinking is the process of coming up with the best answer to a question using our memory, resources around us, or logic.

2. Divergent Thinking

- Divergent thinking is the exact opposite of convergent thinking.
- It involves coming up to solutions, paths forward or new ideas when there is no single correct answer.

4 Types of Thinking Skills

3. Critical Thinking Skills

- Critical thinking skills involve analyzing something in order to form a judgement about it

4. Creative Thinking

- Creative thinking involves thinking about a topic in unusual, unconventional and alternative ways to generate new ideas about an established topic.
- A creative thinker will try to address an issue from a perspective that hasn't been used before.



How To Improve Your Thinking Skills

- There is a difference between types of thinking and methods of thinking.
- Some of the methods include using all of the above types in the form of:
 - Writing
 - Studying
 - Discussing
 - Debating
 - Meditating



REFERENCE

- **Research Methodology** Methods and Techniques (Third Edition) – **C.R.Kothari**

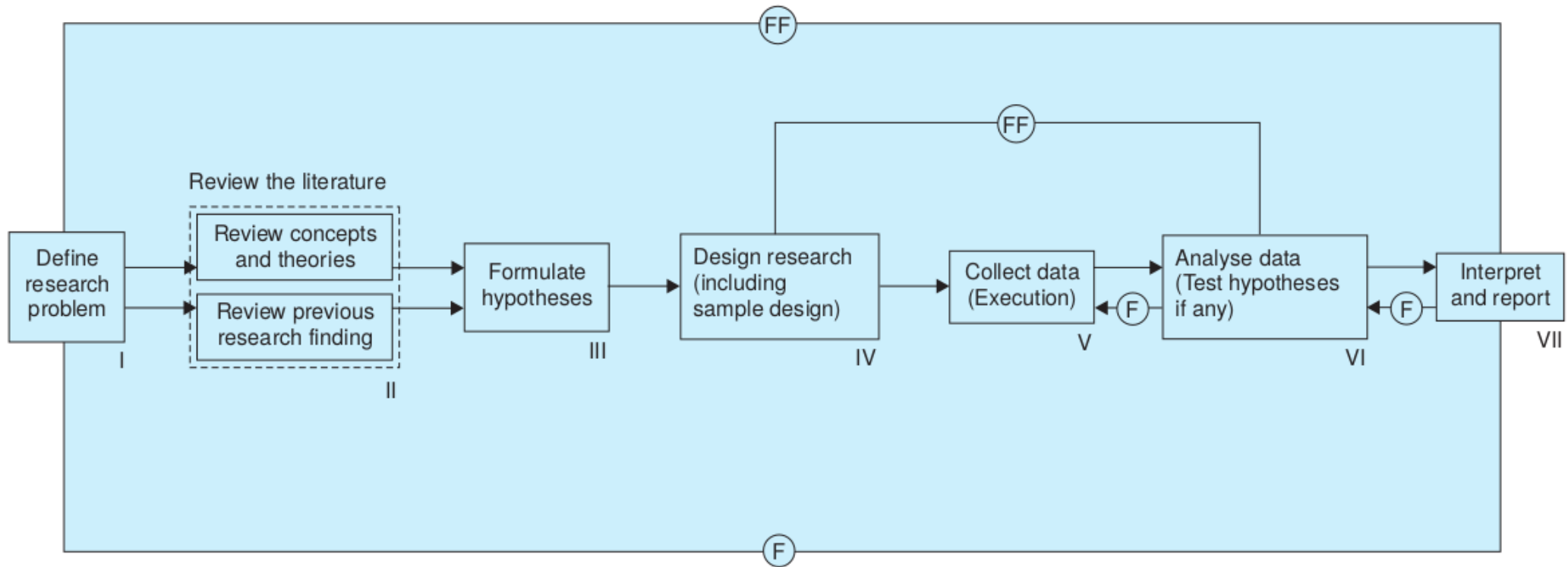
Research Process

The following order concerning various steps provides a useful procedural guideline regarding the research process:

1. formulating the research problem
2. extensive literature survey
3. developing the hypothesis
4. preparing the research design
5. determining sample design
6. collecting the data
7. execution of the project
8. analysis of data
9. hypothesis testing
10. generalisations and interpretation
11. preparation of the report or presentation of the results, ie formal write-up of conclusions reached.

Research Process

RESEARCH PROCESS IN FLOW CHART



Where (F) = feed back (Helps in controlling the sub-system to which it is transmitted)

(FF) = feed forward (Serves the vital function of providing criteria for evaluation)

Research Process

1. Formulating the research problem:

- Initially the problem may be stated in a **broad general** way and then the ambiguities, if any, relating to the problem be resolved.
- The formulation of a general topic into a specific research problem, thus, constitutes the first step in a scientific enquiry.
- Essentially two steps are involved in formulating the research problem-
 - understanding the problem thoroughly, and
 - rephrasing the same into meaningful terms from an analytical point of view.

Research Process

1. Formulating the research problem:

- The best way of understanding the problem is to discuss it with one's own colleagues or with those having some expertise in the matter.
- The researcher must at the same time examine all available literature to get himself acquainted with the selected problem.
- He may review two types of literature —
 - The **conceptual literature** concerning the concepts and theories, and
 - the **empirical literature** consisting of studies made earlier which are similar to the one proposed.

Research Process

2. Extensive Literature Survey:

- The researcher should undertake extensive literature survey connected with the problem.
- For this purpose, the indexing journals and published or unpublished bibliographies are the first place to go to.
- Academic journals, conference proceedings, government reports, books etc., must be tapped depending on the nature of the problem.
- In this process, it should be remembered that one source will lead to another.
- The earlier studies, if any, which are similar to the study in hand should be carefully studied.
- A good library will be a great help to the researcher at this stage.

Research Process

3. Development of working hypothesis

- After extensive literature survey, researcher should state in clear terms the working hypothesis or hypotheses.
- Working hypothesis is tentative assumption made in order to draw out and test its logical or empirical consequences.
- Hypothesis should be very specific because it has to be tested.
- The role of the hypothesis is to guide the researcher by delimiting the area of research and to keep him on the right track.
- It sharpens his thinking and focuses attention on the more important facts of the problem.
- It also indicates the type of data required and the type of methods of data analysis to be used

Research Process

3. Development of working hypothesis

- How does one go about developing working hypotheses?
- The answer is by using the following approach:
 - (a) Discussions with colleagues and experts about the problem, its origin and the objectives in seeking a solution
 - (b) Examination of data and records, if available, concerning the problem for possible trends, peculiarities and other clues
 - (c) Review of similar studies in the area or of the studies on similar problems
 - (d) Exploratory personal investigation which involves original field interviews on a limited scale with interested parties and individuals with a view to secure greater insight into the practical aspects of the problem.
- Thus, working hypotheses arise as a result of a-priori thinking about the subject, examination of the available data and material including related studies and the counsel of experts and interested parties.

Research Process

4. Preparing the research design

- The researcher will be required to prepare a research design, i.e., he will have to state the **conceptual structure** within which research would be conducted.
- The function of research design is to provide for the collection of relevant evidence with minimal expenditure of effort, time and money.
- There are several research designs, such as, **experimental** and **non-experimental** hypothesis testing.
- **Experimental** designs can be either **informal** designs (such as before-and-after without control, after-only with control, before-and-after with control) or **formal** designs (such as completely randomized design, randomized block design, Latin square design, simple and complex factorial designs), out of which the researcher must select one for his own project

Research Process

4. Preparing the research design

- The preparation of the research design, appropriate for a particular research problem, involves usually the consideration of the following:
 - (i) the means of obtaining the information
 - (ii) the availability and skills of the researcher and his staff (if any)
 - (iii) explanation of the way in which selected means of obtaining information will be organised
 - (iv) the time available for research
 - (v) the cost factor relating to research, i.e., the finance available for the purpose

Research Process

5. Determining sample design:

- The researcher must decide the way of selecting a sample or what is popularly known as the sample design.
- A sample design is a definite plan determined before any data are actually collected for obtaining a sample from a given population.
- Samples can be either **probability** samples or **non-probability** samples.
- With **probability** samples each element has a known probability of being included in the sample but the **non-probability** samples do not allow the researcher to determine this probability.
- **Probability** samples are those based on simple random sampling, systematic sampling, stratified sampling, cluster/area sampling whereas **non-probability** samples are those based on convenience sampling, judgement sampling and quota sampling techniques.

Research Process

5. Determining sample design:

- **Deliberate sampling:** **Deliberate** sampling is also known as **purposive** or **non-probability** sampling. When population elements are selected for inclusion in the sample based on the ease of access, it can be called convenience sampling. On the other hand, in judgement sampling the researcher's judgement is used for selecting items which he/she considers as representative of the population.
- **Simple random sampling:** This type of sampling is also known as **chance** sampling or **probability** sampling where each and every item in the population has an equal chance of inclusion in the sample and each one of the possible samples, in case of finite universe, has the same probability of being selected.
- For example, if we have to select a sample of 300 items from a universe of 15,000 items, then we can put the names or numbers of all the 15,000 items on slips of paper and conduct a lottery.

Research Process

5. Determining sample design:

- **Systematic sampling:** In some instances the most practical way of sampling is to select every 15th name on a list, every 10th house on one side of a street and so on. Sampling of this type is known as **systematic** sampling.
- **Stratified sampling:** In this technique, the population is stratified into a number of non-overlapping subpopulations or strata and sample items are selected from each stratum. If the items selected from each stratum is based on simple random sampling the entire procedure, first stratification and then simple random sampling, is known as **stratified random** sampling.
- **Quota sampling:** This is one of the most common forms of **non-probability** sampling. Sampling is done until a specific number of units (quotas) for various sub-populations have been selected.

Research Process

5. Determining sample design:

- **Cluster sampling** involves grouping the population and then selecting the groups or the clusters rather than individual elements for inclusion in the sample.
- **Area sampling** is quite close to cluster sampling and is often talked about when the **total geographical area** of interest happens to be big one. Under area sampling we first divide the total area into a number of smaller non-overlapping areas, generally called **geographical clusters**, then a number of these smaller areas are randomly selected, and all units in these small areas are included in the sample.
- **Multi-stage sampling:** This is a further development of the idea of cluster sampling. This technique is meant for big inquiries extending to a considerably large geographical area like an entire country. Under multi-stage sampling the first stage may be to **select large primary sampling units such as states, then districts, then towns and finally certain families within towns**. If the technique of random-sampling is applied at all stages, the sampling procedure is described as **multi-stage random** sampling.

Research Process

6. Collecting the data:

- Primary data can be collected either through **experiment** or through **survey**. If the researcher conducts an **experiment**, he observes some quantitative measurements, or the data, with the help of which (s)he examines the truth contained in her/his hypothesis.
- But in the case of a **survey**, data can be collected by any one or more of the following ways:
 - (i) **By observation:** This method implies the collection of information by way of investigator's own observation, **without** interviewing the respondents. The information obtained relates to what is currently happening and is not complicated by either the past behaviour or future intentions or attitudes of respondents. The information provided by this method is also very limited.
 - ii) **Through personal interview:** The investigator follows a rigid procedure and seeks answers to a set of pre-conceived questions through personal interviews. This method of collecting data is usually carried out in a structured way where output depends upon the ability of the interviewer to a large extent.

Research Process

6. Collecting the data:

(iii) **Through telephone interviews:** This method of collecting information involves contacting the respondents on telephone itself. This is not a very widely used method but it plays an important role in industrial surveys in developed regions, particularly, when the survey has to be accomplished in a **very limited time**.

(iv) **By mailing of questionnaires:** Questionnaires are mailed to the respondents with a request to return after completing the same. It is the most extensively used method in various economic and business surveys. Before applying this method, usually a Pilot Study for testing the questionnaire is conducted which reveals the weaknesses, if any, of the questionnaire. Questionnaire to be used must be prepared very carefully so that it may prove to be effective in collecting the relevant information.

Research Process

6. Collecting the data:

- (v) **Through schedules:** Under this method the enumerators are appointed and given training. They are provided with schedules containing relevant questions. These enumerators go to respondents with these schedules. Data are collected by filling up the schedules by enumerators on the basis of replies given by respondents.
- The researcher should select one of these methods of collecting the data taking into consideration the nature of investigation, objective and scope of the inquiry, financial resources, available time and the desired degree of accuracy.

Research Process

7. Execution of the project:

- Execution of the project is a very important step in the research process.
- The researcher should see that the project is executed in a **systematic manner** and in **time**.
- Steps should be taken to ensure that the survey is under statistical control so that the collected information is in accordance with the pre-defined standard of accuracy.
- If some of the respondents do not cooperate, some suitable methods should be designed to tackle this problem.
- One method of dealing with the **non-response** problem is to make a list of the non-respondents and take a small sub-sample of them, and then with the help of experts vigorous efforts can be made for securing response.

Research Process

8. Analysis of data:

- After the data have been collected, the researcher turns to the task of analysing them.
- The analysis of data requires a number of closely related operations such as **establishment of categories, the application of these categories to raw data through coding, tabulation and then drawing statistical inferences**
- Researcher should classify the raw data into some purposeful and usable categories.
- Coding operation is usually done at this stage through which the categories of data are transformed into symbols that may be tabulated and counted.
- Tabulation is a part of the technical procedure wherein the classified data are put in the form of tables.
- The researcher can analyse the collected data with the help of various statistical measures.

Research Process

9. Hypothesis-testing:

- Do the facts support the hypotheses or they happen to be contrary?
- This is the usual question which should be answered while testing hypotheses.
- Various tests, such as Chi square test, t-test, F-test, have been developed by statisticians for the purpose.
- The hypotheses may be tested through the use of one or more of such tests, depending upon the nature and object of research inquiry.
- Hypothesis-testing will result in either accepting the hypothesis or in rejecting it.

Research Process

10. Generalisations and Interpretation:

- If a hypothesis is tested and upheld several times, it may be possible for the researcher to arrive at **generalisation**, i.e., to **build a theory**.
- If the researcher had **no** hypothesis to start with, he might seek to explain his findings on the basis of some theory. It is known as **interpretation**. The process of interpretation may quite often trigger off new questions which in turn may lead to further researches.

Research Process

11. Preparation of the report or the thesis:

- Finally, the researcher has to prepare the report of what has been done by him.
- Writing of report must be done with great care keeping in view the following:
 - 1) The **layout of the report** should be as follows: (i) the preliminary pages; (ii) the main text, and (iii) the end matter.
- In its preliminary pages, the report should carry title and date followed by acknowledgements. Then there should be a table of contents followed by a list of tables and list of graphs and charts, if any, given in the report.

Research Process

11. Preparation of the report or the thesis:

- The main text of the report should have the following parts:
 - (a) **Introduction:** It should contain a clear statement of the objective of the research and an explanation of the methodology adopted in accomplishing the research. The scope of the study along with various limitations should as well be stated in this part.
 - (b) **Summary of findings:** After introduction, there would appear a statement of findings. If the findings are extensive, then it should be summarised.
 - (c) **Main report:** The main body of the report should be presented in logical sequence and broken-down into readily identifiable sections.
 - (d) **Conclusion:** Towards the end of the main text, researcher should again put down the **results of his research** clearly and precisely. In fact, it is the final summing up.
- At the end of the report, **appendices** should be enlisted in respect of all technical data. **Bibliography**, i.e., list of books, journals, reports, etc., consulted, should also be given in the end. **Index** should also be given specially in a published research report.

Research Process

11. Preparation of the report or the thesis:

- 2) Report should be written in a **concise** and **objective style** in **simple language** avoiding vague expressions such as 'it seems,' 'there may be', and the like.
- 3) **Charts** and **illustrations** in the main report should be used only if they present the information more clearly and forcibly.
- 4) The **various constraints** experienced in conducting research operations may as well be stated.

Thinking Skills

INTRODUCTION

- People think-that's obvious.
- Human are rational beings & our rationality consists in our ability to think & reason.
- It is because of our capacity for better thinking that we are superior to other animals in learning & in making adjustment.
- During most of our waking hours, & even when we are asleep & dreaming, we are thinking, What are you thinking about right now.

DEFINITION OF THINKING

- Thinking is an activity concerning ideas, symbolic in character initiated by a problem or task which the individual is facing, involving some trial & error but under the directing influence of that problem & ultimately leading to a conclusion or solution of the problem.
-----Warren
- Thinking is the organization & reorganization of current learning in the present circumstances with the help of learning & past experiences.
-----Vinacke (1968)
- Thinking is the perceptual relationship which provides for the solution of the problem.
-----Maier

NATURE OF THINKING

- It is essentially a cognitive activity.
- It is always directed to achieve some end or purpose.
- It is described as a problem-solving behavior.
- It is a symbolic activity.
- It is mental exploration instead of motor exploration.
- It can shift very rapidly.
- It is internal activity.

TYPES OF THINKING

1. Perceptual or concrete thinking
2. Conceptual or abstract thinking
3. Creative thinking
4. Logical thinking/ reasoning
5. Problem solving
6. Convergent Vs Divergent thinking

1. Perceptual or Concrete thinking

- It is based on perception.
 - Perception is the process of interpretation of sensation according to one's experience.
 - It is also called concrete thinking as it is carried over the perception of actual or concrete & events.
 - It is one-dimensional & literal thinking which has limited use of metaphor without understanding nuances of meaning.
 - Being the simplest form of thinking, small children are mostly benefitted by this type of thinking.
- 

2. Conceptual Or Abstract thinking

- It does not require the perception of actual objects or events.
- It is also called abstract thinking as it makes the use of concepts or abstract ideas.
- It is superior to perceptual thinking's as it economizes efforts in understanding & helps in discovery & invention.
- It is ability to appreciate nuances of meaning.
- It is multidimensional thinking with ability to use metaphors & hypotheses appropriately.
- Language plays an important part in conceptual thinking.

3. Creative thinking

- It refers to the ability for original thinking, to create or discover something new.
- It is the ability to integrate the various elements of the situation into a harmonious whole to create something novel.
- In other words, cognitive activity directed towards some creative work refers to creative thinking.
- Creative thinkers are great boons to the society as they enrich the knowledge of mankind.
- The creative thinker tries to achieve something new, to produce something original & something unique

4. Logical thinking/Reasoning

- It is the cognitive process of looking for reasons for beliefs, conclusion, actions or feelings.
- It is the process of drawing conclusions based on evidence.
- It is form of controlled thinking in which the thought process is directed consciously towards the solutions of a problem.
- Reasoning is the highest form of thinking to find out causes & predict effects.
- An individual tries to solve a problem by incorporating two or more aspects of his past experience.

- It is classified into;

I. Inductive reasoning:

This is process of reasoning from parts to the whole, from example to generalizations. It is carried out generally within the field known as informal logic or critical thinking.

II. Deductive reasoning:

This moves from the whole to part, from generalization to underlying concepts to examples. Formal logic is described as “the science of deduction.”

III. Abductive reasoning:

It is cognitive process often involves both inductive & deductive arguments.

5. Problem solving

- It is a tool, skill & a process.
- It is a tool because it can help you solve an immediate problem or to achieve a goal.
- It is a skill because once you have learnt it you can use it repeatedly, like the ability to ride a bicycle, add numbers or speak a language.
- It is also a process because it involve taking a number of steps.
- You can engage in problem solving if you want to reach a goal & experience obstacles on the way.
- At the point at which you come up against a barrier you can engage in a problem solving process to help you achieve your goal.

Count...

- Every time you use a problem solving process, you are increasing your problem solving skill.
- **A seven-step problem solving cycles;**
 1. Identify the problem
 2. Explore the problem
 3. Set goals
 4. Look at alternatives
 5. Select a possible solution
 6. Implement a possible solution
 7. evaluation

6. Convergent Vs Divergent thinking

- Convergent thinking is cognitive processing of information around a common point, an attempt to bring thoughts from different directions into a union for common conclusion
- Divergent thinking starts from a common point & moves outward into a variety of perspectives. Eg; teachers use the content as a vehicle to prompt diverse or unique thinking among students rather than a common view.

LEVELS OF THINKING

- There are six levels of thinking / learning within cognitive hierarchy of behaviors;
 1. Level 1: Knowledge
 2. Level 2: Comprehension
 3. Level 3: Application
 4. Level 4: Analysis
 5. Level 5: Synthesis
 6. Level 6: Evaluation

LEVEL1: KNOWLEDGE

- It involve recall of facts, principles & terms in the forms in which they are learned.
- *Key words:* who, what, when, omit, where, which, choose, find, how, define, label, show, spell, list, match, name, relate, tell, recall, select.
- Eg; When did -----happen?
Which one -----?

LEVEL 2: COMPREHENSION

- It represent the lowest form of understanding.
- The student knows what is being communicated without relating it to other material or seeing it in its fullest meaning.
- It demonstrates understanding of facts & ideas by organizing, comparing, translating, interpreting, giving descriptions & stating main ideas.
- *Key words:* Compare, contrast, demonstrate, interpret, explain, extend, illustrate, infer, outline, relate, rephrase, translate, summarize, show, classify.
- Eg; What facts or ideas show-----?
Which is the best answer----- ?

LEVEL 3: APPLICATION

- It involve the use of abstractions in concrete situation such as nursing or other specific situations.
- It is solving problems by applying acquired knowledge, facts, techniques & rules in a different way.
- Key words: apply, build, choose, construct develop, interview, make use of, organize, experiment with plan, select, solve, utilize, model, identify.
- Eg; What would result if-----?

Can you make use of the facts to-----?

LEVEL 4: ANALYSIS

- It is examining & breaking information into parts by identifying motives or causes; making inferences & finding evidence to support generalization.
- *Key words:* analyze, categorize, compare, contrast, discover, dissect, divide, examine, inspect, simplify, survey, take part in, test for, distinguish, list, distinction, theme, relationships, function, motive, inferences, assumption, conclusion.
- Eg; What are key parts or features of-----?
What motive is there-----?
What evidence can you find-----?

LEVEL 5: SYNTHESIS

- It is compiling information together in a different way by combining element in a new pattern or proposing alternative solution.
- *Key words:* build, choose, combine, compile, compose, construct, create, design, develop, estimate, formulate, imagine, invent, makeup, originate, plan, predict, propose, solve solution etc.
- Eg; Can you propose an alternative-----?
Can you formulate a theory for-----?
What facts can you compile-----?

LEVEL 6: EVALUATION

- It is presenting & defending opinion by making judgment about information, validity of ideas or quality of work based on a set of criteria.
- Key words: award, choose, conclude, criticize, decide, defend, determine, evaluate, judge, justify, measure, compare, mark, rate, recommend, prioritize, prove, disprove etc.
- Eg; What is your opinion of-----?
Would it be better if-----?
How would you rate the-----?

WAY TO IMPROVE THINKING

1. **Keep yourself only with current project:**

- Clutter creates confusion.
- Get organized & work with a fresh canvas.
- On an average, we spend about 45 minutes a day looking for things eg; Where are my keys?
- Keep only the current project you are working on in front of you.

2. Dedicate an hour of focus time to your most important task:

- Multi-tasking is highly over-rated & causes a loss of upto 40% efficiency.
- Get yourself one hour of focused seclusion to work on your important task.
- It will increase both productivity, creativity & morale.

3. Stir up your visual & creative talents with exposure to the arts:

- Go to a gallery, pick up an art book or spend time with nature.

4. Learn how to ‘mind map’:

- This is a best practice that allows you to visualize & map your projects, & strategies.
- It is also a life saving memory device that will help you remember more & organize your thinking.

5. Finally, on the hour-----stop thinking

- When you have been working on something for more than an hour, you start losing concentration & focus.
- So, stop, get up & walk around & then go back to your work.

THINKING IN RELATION TO LANGUAGE AND COMMUNICATION

- Through the use of language, we can develop complex & abstract concepts. We learn through discussion the essential characteristics of concepts.
- Language like thought is directed to some purpose or goal. It has a unity & organization about the goal. Language is the symbolic vehicle by which thought is carried.
- Language allow new learning to be communicated to others & saved for future generations.
- The use of language – the communication of information.
- Language provides us with categories that we use to construct our view of people & events in the world around us.

RESEARCH

Scientific thinking

- ◉ Systematic
- ◉ Proper analysis present
- ◉ Controlled conditions
- ◉ Empirical testing
- ◉ Conscious Pursuit of relationship
- ◉ No metaphysical explanations
- ◉ 2 or more variables
- ◉ Using this research is carried out

Common sense thinking

- ◉ Formed loosely
- ◉ Proper analysis absent
- ◉ No control over conditions
- ◉ Fanciful explanations
- ◉ Not accurate
- ◉ Layman's terms
- ◉ No experiments
- ◉ No variable control
- ◉ Motivates research [gives topic for research]

Characteristics of scientific thinking

- Systematically building of theoretical concepts.
- It involves testing hypothesis systematically, empirically in controlled settings.
- We have 2 main types of variables: training and academic performance.
- Involves conscious and systematic pursuit of relationship.
- Rules out metaphysical explanations: soul, mind, destiny

Characteristics of common sense thinking

- Involves implicit concepts.
- Concepts are loosely formed.
- Makes us accept fanciful experiences.
- Formation of hypothesis is subjective.
- Rarely thinking of controlling variables(statements based on prejudice).
- Two phenomenon are linked as cause and effect just because of co-occurrence.

Division into sub-problems

- Main research problem is divided into sub problems that have can be answered with data collection or research.
- Each sub problem is asked as a question, answered by a hypothesis, which is tested and accepted or rejected via defined data plan.
- Main research problem includes words and direct relevance to sub problems.

Dividing the Research Problem into Subproblems

- Most research problems are too large, broad, or complex.
- **The Solution:** Break down into sub-problems
- It helps the researcher with **HOW** to implement a strategy in the research process

4 Characteristics of Subproblems

- **Must be researchable**
- **Must be tied to the interpretation of data**
- **Subproblems must add up to the total problem**
- **No more than 2 to 6 subproblems (3 is the most common)**

Hypothesis

- A tentative proposition to answer the subproblem question and guide the investigative process
- Generally, there is a 1:1 ratio of hypothesis to subproblem
- Not proven or disproven, **but supported or not supported**
- Make a declarative statement

Research Plan or Map

- Sometimes it is best to decide what you want to cover first and use the sub-problems as the starting point
- Develop hypothesis(es) for each sub-problem
- Determine the criteria needed to support (or not) the hypothesis
- Design an “Umbrella” or Main Research Question that encompasses all sub-problems

Basic Research

- Collecting information from peer reviewed articles, journals, encyclopaedias, dissertations, studies, mentors library collection etc..enhances ones conceptual knowledge to topic
- Each research project needs at least one basic research sub problem to demonstrate depth of knowledge in the field of study.

Applied Research

Actively collecting data from an experiment, survey, observation, etc.

Each Research Project should contain a subproblem that has relevance to the actual internship experience.

Examples: Opinions of medical experts on a topic, surveys of employees, data from sales of concrete, data of most favored aroma, steps involved in a particular surgery

Sometimes a sub-problem can incorporate both basic and applied research.

A good research project should have BOTH Applied and Basic sub-problems.

Pseudoproblems

- Do not confuse subproblems with **pseudoproblems**
- **Pseudoproblems** describe more the procedure of the study and not the problem itself.
- Ex: How large should the representative sample be? What statistical tool should be used?

Creativity

Introduction

- Creativity, like intelligence and personality, is a commonly used term in our day-to-day life.
- But do we really understand what constitutes creativity?

What is creativity?

- We all are creative to various extents.
- We have exhibited creative thinking at some point or other when faced with life's varied situations and challenges.
- It can be called as a special kind of problem solving.

Creativity

- Creativity can be described in reference to people, the inventive acts or products, a new creation like a composition or an art/sculpture, or a scientific theory.
- In other words, we mainly refer creativity to the person or to some tangible product or new creation of that individual.

Creativity

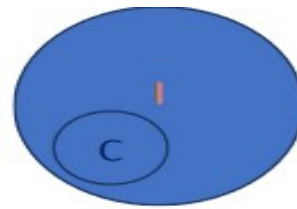
- We also describe the creative persons in terms of various specific characteristics. Thus, creativity can be considered
 - (a) from the point of view of the person who creates
 - (b) in terms of the mental and emotional processes such as perception, motivation, learning, thinking and communication
 - (c) in terms of its products such as inventions, theories, poems, paintings etc.

Intelligence versus Creativity

- Creativity and intelligence relationship has been widely researched and the 'threshold hypothesis' has been prominent.
- It states that high intelligence does not guarantee creativity; but a certain level of intelligence (a particular threshold level) is required in order to be creative.
- Threshold hypothesis for the interplay between intelligence and creative potential.

Intelligence versus Creativity

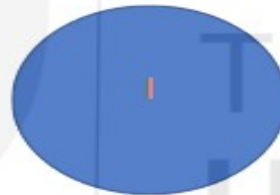
- Once the intelligence threshold is met, personality factors become more predictive for creativity.
- On the contrary, no threshold was found for creative achievement, i.e., creative achievement benefitted from higher intelligence even at fairly high levels of intellectual ability.



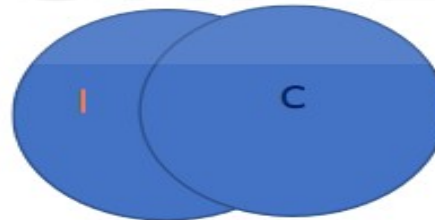
a) Creativity as a part of intelligence



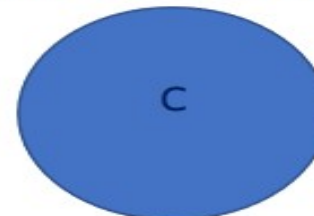
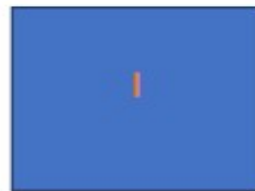
b) Intelligence as a part of creativity



c) Creativity and intelligence viewed as identical sets



d) Creativity and intelligence as independent, but overlapping constructs



e) Creativity and intelligence as completely unrelated sets.

Figure 10bF: Relationship between Creativity and Intelligence

Creative Process

- Creativity is a complex process. It does not come all of a sudden.
- It takes time but creative ideas appear all of a sudden, like a flash of lightning.
- According to Graham Walls (1926), there are four main stages in creativity:
 - **Preparation, Incubation, Inspiration, Verification**
- **Preparation:**
 - The person in this stage tries to understand the problem.
 - As Einstein had said, “the formulation of a problem is often more essential than its solution”.
 - Thus, focus needs to be more on knowing about the problem in detail.
 - It involves setting the goal and generating ideas towards the solution of the problem.

Creative Process

- **Incubation:**

- This is the time out phase where the individual does not engage in any activity.
- The processing of all the information gathered during the preparation stage goes on in this stage.
- The person does not consciously or actively think about the problem, but the creative processes carry on within the individual at an unconscious level.

Creative Process

- **Inspiration:**

- the person suddenly gets an insight about the solution of the problem.
- A new idea or new possibilities emerge and the individual is more active and conscious in this stage.
- It may be noted that this 'eureka' moment involves a lot of thinking and effort in the preparation stage.

- **Verification:**

- The person then tries to evaluate and verify the ideas and solutions in this stage.
 - It may result in some modifications of the solution.
 - In some cases, the person may also need to go back to the first stage to gather or generate more information about the problem, thus more 'preparation' is required.
- The process (4 stages) is cyclical.

Course Outcome

- CO 1: Approach research projects with enthusiasm and creativity.